Course Code	Course Name	Hours Per Week			
		L	T	P	Credits
CS3CO37	Advanced Java Programming	2	0	2	3

Course Learning Objectives (CLOs):

CLO_{01} Have a broad understanding of Java collections and Generic

CLO_{02} Have a high-level understanding of basic programming principles to the construction of websites using servlet

CLO_{03} Be able to design user interactions on web pages, develop back-end website application and databases for functionality

CLO_{04} Be able to design web page on Spring Framework.

CLO_{05} Be able to develop web project using spring boot and JDBC.

Unit-1 Collection and Generic

Introduction to Generics, Generics Types and Parameterized Types, Wildcards, Java Collection Framework, Collections (Basic Operations, Bulk Operations, Iteration) List, Set, Maps Lambda Expressions - Lambda Type Inference, Lambda Parameters, Lambda Function Body, Returning a Value, From a Lambda Expression, Lambdas as Objects.

Unit-2 Introduction Java EE Programming and Servlets

Basics of Web Application, web client and web server, Servlets, HTTP Methods; GET, POST, PUT, DELETE, TRACE, OPTIONS, MVC design pattern, Init Parameters, Servlet Context, Inter Servlet Communication, Servlet Listeners, Servlet Filters.

Unit-3 JDBC and JSP

Managing JDBC Connection, Configuring Data Source to obtain JDBC Connection, Data Access operations with JDBC Template, RDBMS operation classes, JSP Architecture, JSP building blocks, Scripting Tags, implicit object, Introduction to Bean, standard actions, session tracking types and methods. Custom Tags, Introduction to JSP Standard Tag Library (JSTL) and JSTL Tags.

Unit-4 Spring Frameworks

Introduction to Spring Framework, OJO Programming Model, Lightweight Containers (Spring IOC container, Configuration Metadata, Configuring and using the Container) Dependency Injection with Spring- Setter Injection, Constructor Injection

Unit-5 JDBC and Spring Boot

Data Access operations with JDBC Template and Spring , Modelling JDBC Operations as Java Objects, Spring Boot and Database, Spring Boot Web Application Development

Text Books:

1. “Core Java for Impatients”, Cay S. Horstman
2. “Java: The Complete Reference”, Herbert Schildt

References:

1. “Head First Java”, Kathy Sierra, Bert Bates
2. “Java for Dummies”, Barry A. Burd
3. “Effective Java”, Joshua Bloch

Course Outcomes (COs):

After completion of this course the students shall be able to:

CO_{01} Identify the basic concepts collection class and generic method and class.

CO_{02} Apply basic design principles to present ideas, information, products, and services on websites

CO_{03} Apply basic programming principles to the construction of websites and database connectivity.

CO_{04} Apply basic programming principles to the construction of websites using Spring frameworks.

CO_{05} Design and develop Application Programming Interfaces.

List of Practical :

1. Write a Java Program to demonstrate a Generic Class, Generic methods and wildcards.
2. Write a Java program to create List containing list of items of type String and use for-each loop , Iterator interface, ListIterator interface to print the items of the list.
3. Write a Java program using Set interface containing list of items and perform the following operations:
   a. Add items in the set.
   b. Insert items of one set in to other set.
   c. Remove items from the set
   d. Search the specified item in the set
4. Write a Java program using Map interface containing list of items having keys and associated values and perform the following operations:
   a. Add items in the map.
   b. Remove items from the map
   c. Search specific key from the map
   d. Get value of the specified key
   e. Insert map elements of one map in to other map.
   f. Print all keys and values of the map.
5. Write a Java program using Lambda Expression with multiple parameters to add two numbers and to concatenate two strings.
6. Write a JSP page to display the Registration form (Make your own assumptions).
7. Write a JSP program to add, delete and display the records from StudentMaster (RollNo, Name, Semester, Course) table.
8. Write a JSP program that demonstrates the use of JSP declaration, scriptlet,directives
9. Write a JSP program that demonstrates the use of JSP expression, header and footer.
10. Design loan calculator using JSP which accepts Period of Time (in years) and Principal Loan Amount. Display the payment amount for each loan
11. Write a program to demostrate get and post method using servlets?
12. Write a program to implement servlet listeners and servlet filters methods.
13. Write a program to print “Hello World” using spring framework.
14. Write a program to demonstrate dependency injection via setter method.
15. Write a program to demonstrate dependency injection via Constructor.
16. 1. Write a program to demonstrate Spring AOP – before advice, after advice, around advice.
17. 4. Write a program to demonstrate Spring AOP – after returning advice, after throwing advice and pointcuts.
18. Write a program to insert, update and delete records from the given table.
19. Write a program to demonstrate PreparedStatement in Spring JdbcTemplate.
20. Write a program in Spring JDBC to demonstrate ResultSetExtractor Interface.
21. Write a program to demonstrate RowMapper interface to fetch the records from the database.
22. Write a program to create a simple Spring Boot application that prints a message.